HK IMO Training 2023-2024
Problem Set 7
For 18 November 2023

Problem 25. In $\triangle ABC$, let M be the midpoint of the arc BC of the circumcircle of $\triangle ABC$ that does not contain A. Let ℓ be the line passing through M and parallel to AB. Suppose $D \neq B$ is a point on the line AB such that the circumcircle of $\triangle BDM$ is tangent to ℓ. Prove that $AC = AD$.

Problem 26. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f\big(f(x)f(1-x)\big) = f(x) \quad \text{and} \quad f(f(x)) = 1 - f(x)$$

for all $x \in \mathbb{R}$.

Problem 27. Prove that there are infinitely many odd composite positive integers n such that $n \mid 2^{\frac{n-1}{2}} + 1$.

Problem 28. Let n be a positive integer, and let $S = \{1, 2, \ldots, n\}$. For each of the 2^n subsets of S, we colour the subset black or white. Then for any subset T of S, we denote by $f(T)$ the number of subsets of T which are black. Find the number of ways to colour the subsets such that

$$f(T_1)f(T_2) = f(T_1 \cup T_2)f(T_1 \cap T_2)$$

for any subsets T_1 and T_2 of S.